



NATIONAL TEXTILE UNIVERSITY

Department of Computer Science

Lab Plan # 4: Programming Fundamentals (COC-1071)

Basic Information			
Registration#		Name	
Total Marks		Marks Obtained	
Tools:	Dev-C++ 5.4.1		
Objectives	1. While Loop	2. For Loop	
Note	Solve the following problems using the concepts we have covered so far		

While Loop Questions:

1. Write a C++ program to find the sum of all even numbers from 1 to 100 using a while loop.
2. Create a C++ program that calculates the factorial of a given positive integer using a while loop.
3. Implement a C++ program to reverse a given number using a while loop.
4. Write a C++ program to check if a given number is prime or not using a while loop.
5. Develop a C++ program that computes the Fibonacci sequence up to a specified number of terms using a while loop.
6. Write a C++ program to find all the prime numbers within a specified range using a while loop.
7. Write a C++ program to count the number of digits in a positive integer using a while loop.
8. Implement a C++ program that calculates the average of a list of numbers using a while loop.
9. Create a C++ program to find the square root of a given number using the Newton-Raphson method and a while loop.
10. Develop a C++ program that checks if a given number is a palindrome using a while loop.

For Loop Questions:

11. Write a C++ program to print the multiplication table of a given number using a for loop.
12. Create a C++ program that calculates the sum of the first N natural numbers using a for loop.
13. Implement a C++ program to find the factorial of a given number using a for loop.
14. Write a C++ program to print the pattern of an inverted right-angled triangle using a for loop.
15. Develop a C++ program that checks if a given number is a perfect number using a for loop.
16. Create a C++ program to calculate the exponentiation of a number using a for loop.
17. Write a C++ program to find the greatest common divisor (GCD) of two numbers using a for loop.
18. Implement a C++ program that prints the first N prime numbers using a for loop.
19. Create a C++ program to calculate the sum of the digits of a given number using a for loop.
20. Develop a C++ program that generates the Fibonacci sequence up to a specified number of terms using a for loop.

Shapes Questions:

1. **Hollow Square Pattern:** Write a C++ program to construct a hollow square pattern of asterisks (*) with a specified side length. For example, if the side length is 5, the pattern would look like this:

```
* * * * *
*       *
*       *
*       *
*       *
* * * * *
```

2. **Right Triangle Pattern:** Create a C++ program to construct a right triangle pattern of asterisks (*) with a specified height. For example, if the height is 4, the pattern would look like this:

```
*
* *
* * *
* * * *
```

3. **Pyramid Pattern:** Implement a C++ program to construct a pyramid pattern of asterisks (*) with a specified height. For example, if the height is 5, the pattern would look like this:

```
  *
 * *
* * *
* * * *
* * * * *
```